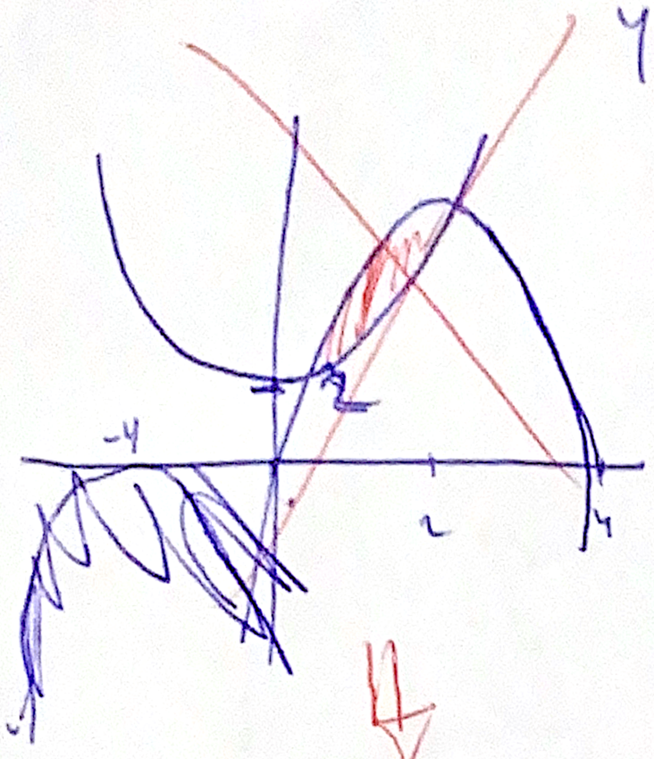
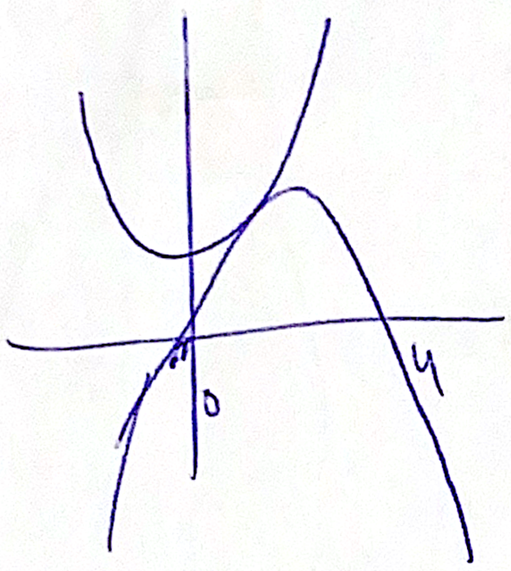
INT1709 - Guía 2

5.

e) $y = x^2 + 2$ e $y = 4x - x^2$

$y = x(4 - x)$ } $x = 0$; $x = 4$





Intersecciones:

(igualamos las dos funciones)

$x^2 + 2 = 4x - x^2$

$2x^2 - 4x + 2 = 0$

$x^2 - 2x + 1$

$(x - 1)^2 = 0$

$1 = x$ } 1 intersección ⇒ Tangentes

0 soluciones ⇒ no hay área en ambas

f) $y = x^2$

$y = 8 - x^2$

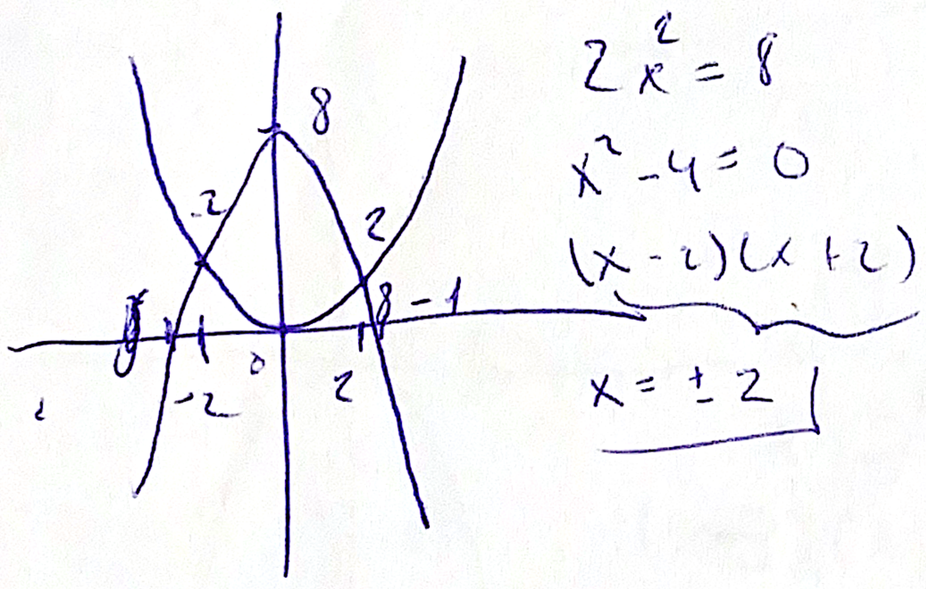
$x^2 = 8 - x^2$

$2x^2 = 8$

$x^2 - 4 = 0$

$(x - 2)(x + 2)$

$x = \pm 2$



$A = 2\int_0^2 F(B) - F(A)\, dx$

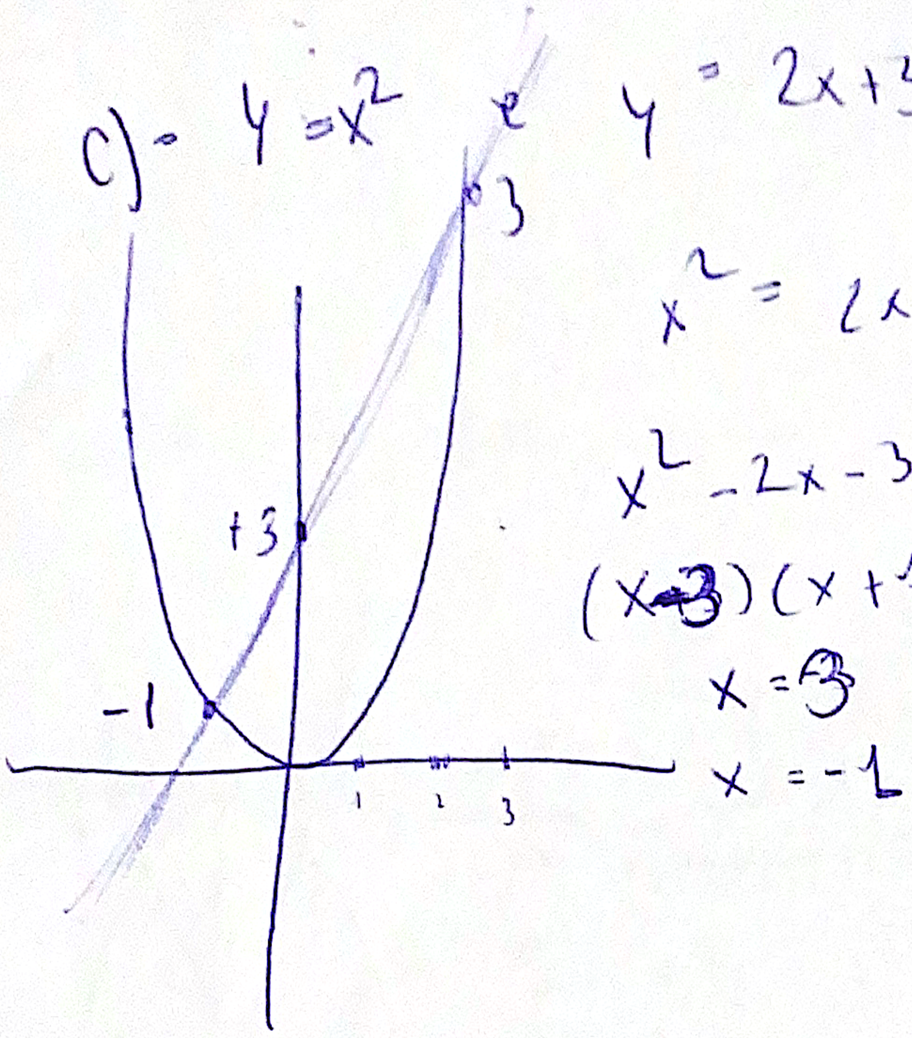
$A = 2\left(\left(-\frac{x^3}{3}\right) \cdot 2 + 8x\right)\Big|_0^2 = 2\left(16 - \frac{16}{3}\right) - (0) \cdot 2$

$= \left(32 - \frac{32}{3}\right) - 0$

$= \frac{96 - 32}{3} = \frac{64}{3}$ u²

$\therefore \frac{16 \cdot 6}{96}$

c) - $y = x^2$ e $y = 2x + 3$



$x^2 = 2x + 3$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1)$

$x = 3$

$x = -1$

$$= \int_{-1}^{3} (2x + 3 - x^2)\,dx$$

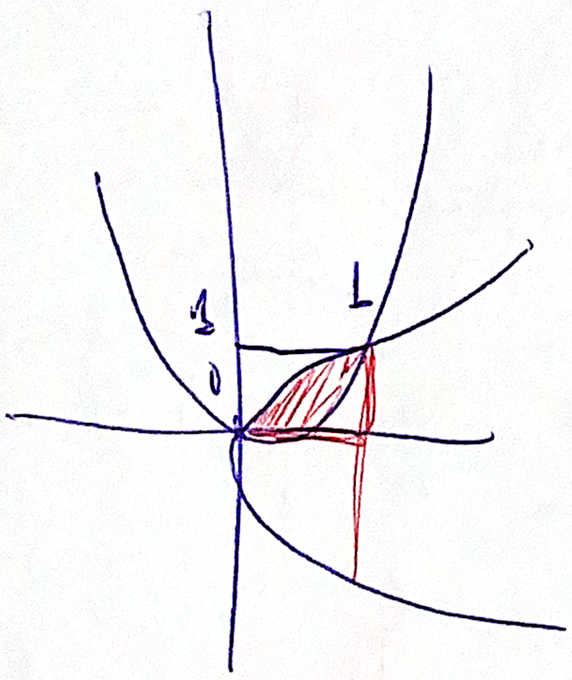
$$= x^2 + 3x - \frac{x^3}{3}\Big|_{-1}^{3}$$

$$= \left(9 + 9 - 9\right) - \left(1 + -3 - \frac{1}{3}\right)$$

$$9 - \left(-2 - \frac{1}{3}\right) = 9 - \frac{11}{3}$$

$$9 + \frac{7}{3} = \frac{34}{3}$$

$y = \sqrt{x}$ $y = x^2$



$x^4 - x = 0$

$x(x^3 - 1) = 0$

$x = 0$ $x = 1$

$$V = \pi \int_0^1 \left[(\sqrt{x})^2 - x^4\right] dx = \frac{x^2}{2} - \frac{x^5}{5}\Big|_0^1$$

$$= \left(\frac{1}{2} - \frac{1}{5}\right) - (0)$$

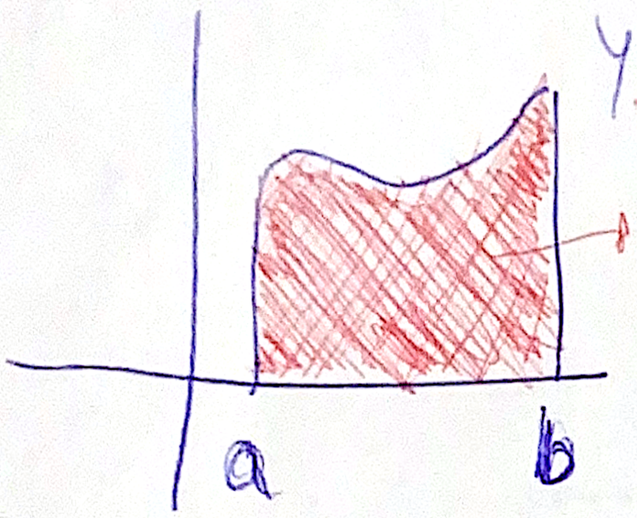
$$= \frac{3}{10}\ u^3$$

La integral definida:

Motivación (el problema del área bajo la curva)

Consideremos la función $f:[a,b] \to \mathbb{R}$ (codominio) en primer positivo.

(nombre) definida en $[a,b]$ → entrega reales.



Para calcular integrales definidas:

1) Suma de Riemann: Corresponde a la suma del área de los (n) rectángulos [dan + o del área bajo la curva].

n altura

$$\sum_{i=1} \overbrace{f(x_i)}^{\text{altura}} \underbrace{\Delta x_i}_{\text{base}}$$

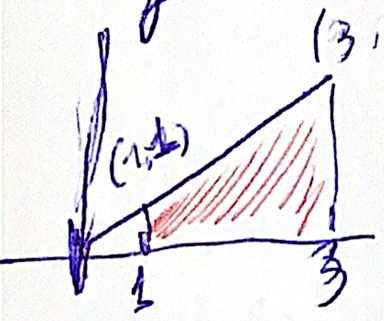
2) Integral definida: $\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x_i = \int_a^b f(x)\,dx$ → Diferencial de x: Δx que tiende a 0.

Teorema fundamental del cálculo. (2da forma):

Si $F(x)$ es una antiderivada de $f(x)$ ~~en~~ y f es continua en $[a,b]$

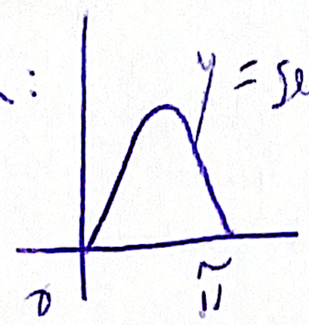
entonces: $\int_a^b f(x)dx = F(b) - F(a) := F(x)\Big|_a^b$ (notación)

Ej: $f(x) = x$



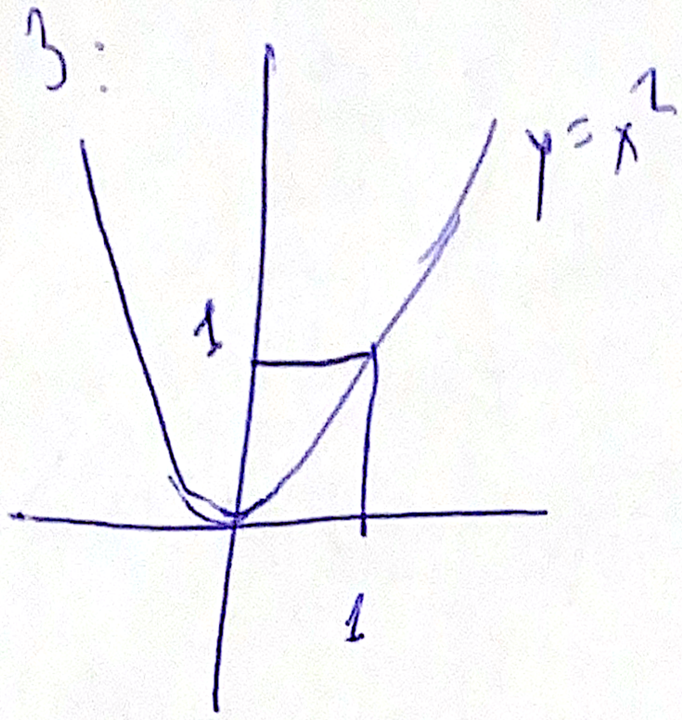
$$A = \int_1^3 x\,dx = \frac{x^2}{2}\Big|_1^3 = \frac{9}{2} - \frac{1}{2} = \frac{4}{1}$$

Ej2:



$$A = \int_0^{\pi} \operatorname{sen}(x)\,dx = -\cos(x)\Big|_0^{\pi} = 1 - (-1) = 2$$

Ejemplo 3:



$$\int_0^1 x^2 dx = \left(\frac{x^3}{3} + C\right)\Big|_0^1 = \left(\frac{1}{3} + C\right) - (C)$$

$$= \frac{1}{3} [u^2]$$

área en unidades cuadradas

Ejemplo 4:

$$\int_0^1 \sqrt{1 - x^2}\, dx$$ ta

$x = \operatorname{sen}(\theta)$

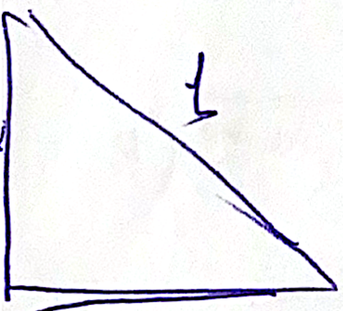
$dx \cos\theta\, d\theta$

$\operatorname{sen}(\alpha + \beta) = \operatorname{sen}\alpha\cos\beta + \operatorname{sen}\beta\cos\alpha$

$$\int \sqrt{1 - \operatorname{sen}^2}\cos\theta\, d\theta = \int \cos^2(\theta)\, d\theta = \int \left(\frac{\cos(2\theta)}{2} + \frac{1}{2}\right) d\theta = \frac{\theta}{2} + \frac{\operatorname{sen}(2\theta)}{4}$$

$$= \frac{\theta}{2} + \frac{2\operatorname{sen}\cos(\theta)}{4}$$

$$\int_0^1 \left(\frac{1}{2}\arcsen(x) + \underbrace{\frac{x\sqrt{1 - x^2}}{2}}_{0}\right)\Big|_0^1$$

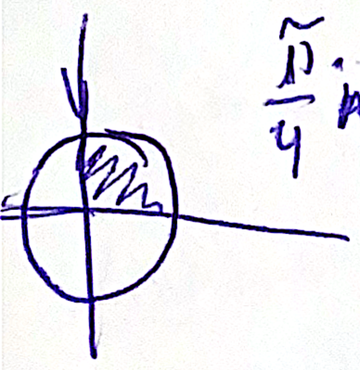


$$= \frac{\arcsen(x)}{2} + \frac{x\sqrt{1 - x^2}}{2}$$

$$= \frac{\arcsen(1)}{2} - \frac{\arcsen(0)}{2} = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

$x^2 + y^2 = 1$

$y = \pm\sqrt{1 - x^2}$



$$\int_{-3}^{-1} x\, dx = \frac{x^2}{2}\Big|_{-3}^{-1}$$

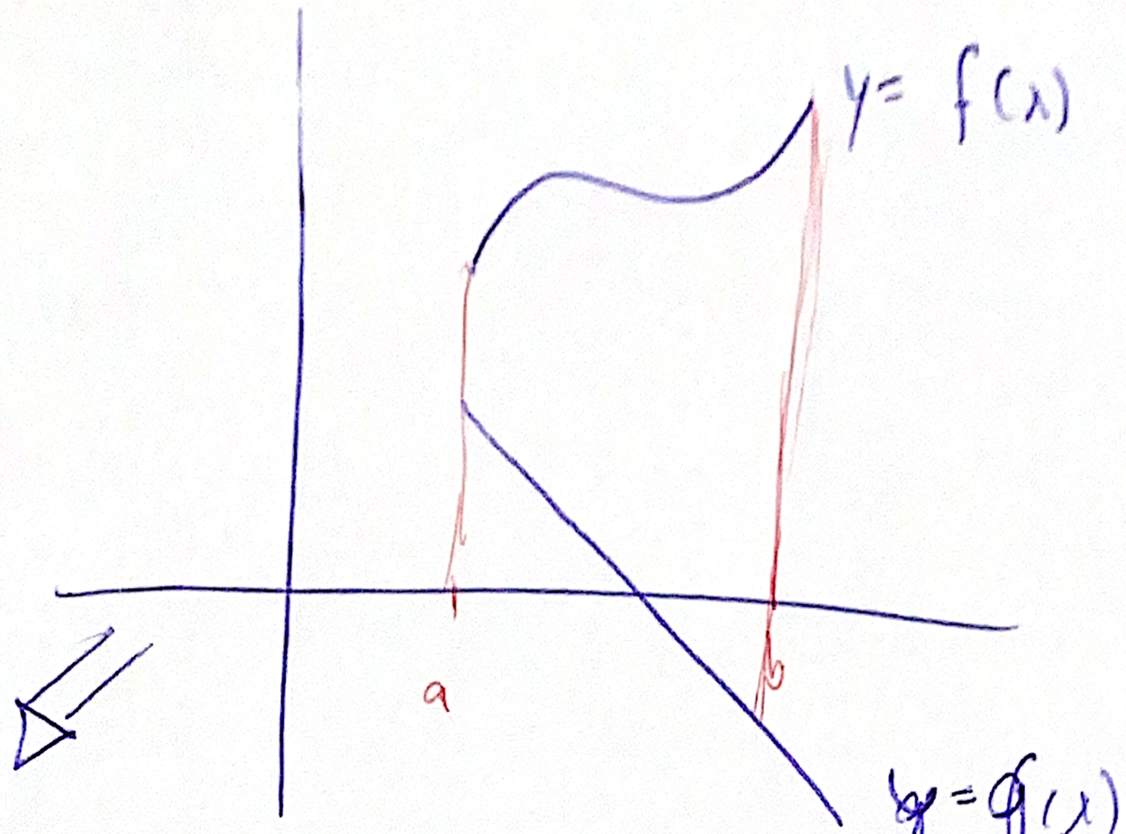
$$\int_{-1}^{1} x\, dx = \frac{x^2}{2}\Big|_{-1}^{1}$$

$$= \frac{1}{2} - \frac{1}{2} = 0$$ X

$\therefore \int_a^b$ debe tener $a > 0$, $b > 0$ o $a < 0$, $b < 0$

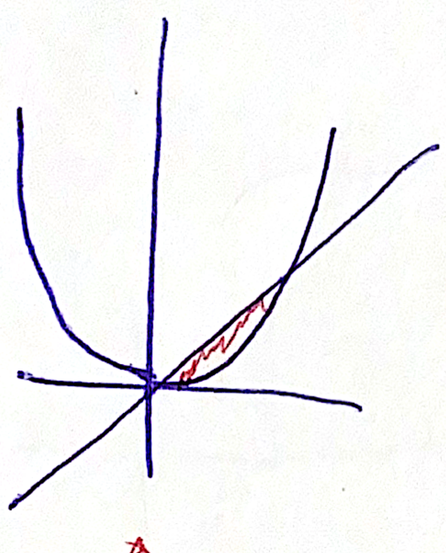
Área entre curvas: Sean $f, g : [a,b] \to \mathbb{R}$ tales que $f(x) \geq g(x)$. Entonces el área comprendida entre las curvas $y = f(x)$ e $y = g(x)$ es:

$$A = \int_a^b (f(x) - g(x))\,dx$$



$$A = \int_a^b (f(x) - g(x))\,dx$$

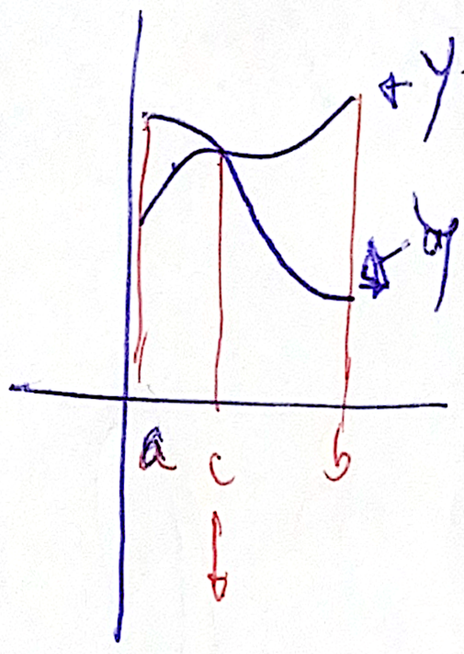
Ej: Encontrar el área de la región más acotada comprendida entre las curvas $y = x$, $y = x^2$



$$A = \int_0^1 (x - x^2)\,dx$$

$$A = \left(\frac{x^2}{2} - \frac{x^3}{3}\right)\Big|_0^1$$

$$= \left(\frac{1}{2} - \frac{1}{3}\right) - (0 - 0) = \frac{1}{6}\ u^2$$



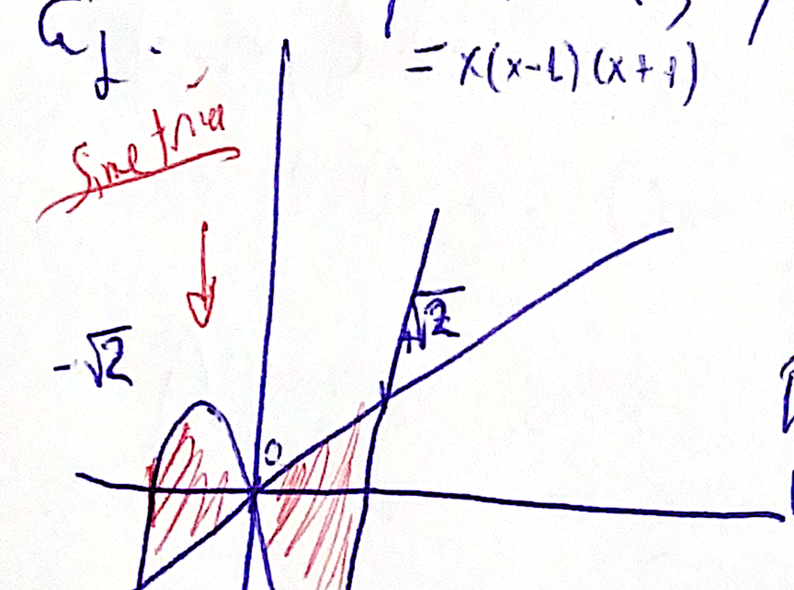
punto de intersección.

$$= \int_a^c (g(x) - f(x))\,dx + \int_c^b (f(x) - g(x))\,dx$$

Ej: Hallar el área encerrada

$y = x^3 - x$, $y = -x$

$= x(x-1)(x+1)$



$$\int_{-\sqrt{2}}^{0} ((x^3 - x) - (-x))\,dx + \int_0^{\sqrt{2}} ((-x) - (x^3 - x))\,dx = \left(\frac{x^4}{4} - \frac{x^2}{2}\right) - \left(\frac{x^2}{2}\right)\Big|_{-\sqrt{2}}^{0}$$

$$+ \frac{x^2}{2} - \left(\frac{x^4}{4} - \frac{x^2}{2}\right)\Big|_0^{\sqrt{2}}$$

Puntos de intersección: $x^3 - x = -x$

$(x^2 - 2)x = 0$

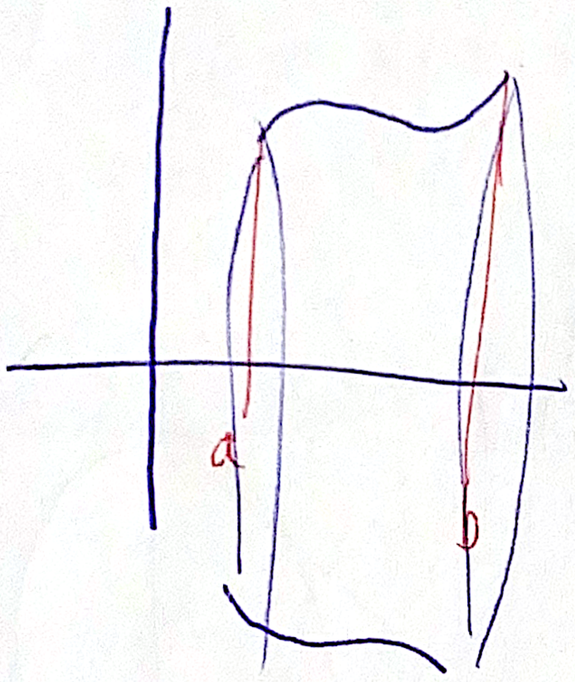
$(x + \sqrt{2})(x - \sqrt{2})\,x = 0$

$$= \frac{x^4}{4} - x^2\Big|_{-\sqrt{2}}^{0} + x^2 - \frac{x^4}{4}\Big|_0^{\sqrt{2}}$$

$$= 0 - (1 - 2) + (2 - 1) + 0 = 1 + 1 = 2$$

Volumen de sólidos de revolución: Sea $f:[a,b]\to\mathbb{R}$

Si dicha (región) rota entorno al eje X, entonces su volumen está dado por:

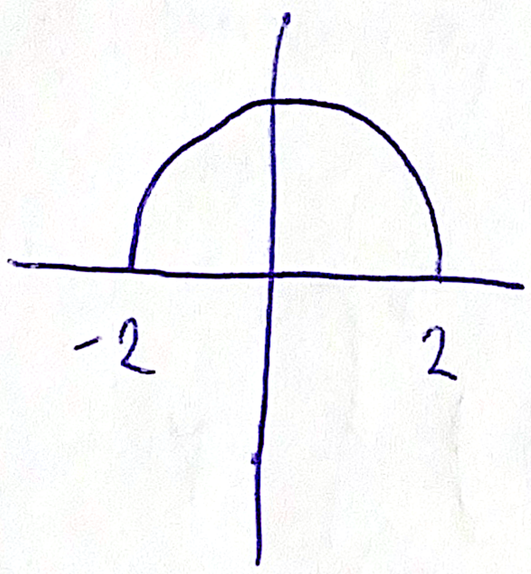


$$V=\int_a^b \pi\left(f(x)\right)^2 dx = \pi\int_a^b \left(f(x)\right)^2 dx$$

solo método del disco

[para: ⊙ y:

Ej 2: $y=\sqrt{4-x}$ entre -2 y 2



$$V=\pi\int_{-2}^{2}(4-x^2)\,dx$$

$$=\pi\left(4x-\frac{x^3}{3}\right)\Big|_{-2}^{2}$$

$$V_4=\frac{4}{3}\pi r^3=\frac{32\pi}{3}$$

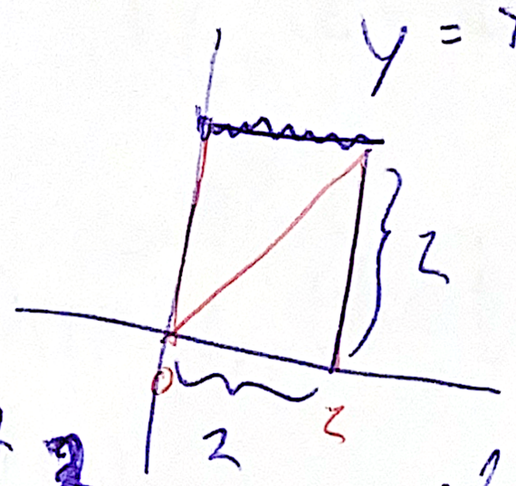
$$=2\pi\left(4x-\frac{x^3}{3}\right)\Big|_0^2$$

$$=2\pi\left(8-\frac{8}{3}\right)-(0-0)$$

$$=2\pi\left(\frac{16}{3}\right)=\frac{32\pi}{3}$$

entre $x=0$ y $x=2$

$y=x$



$$\therefore V=\pi\int_0^2 (x)^2\,dx \quad \therefore V=\pi\int_0^2 (x)^2\,dx$$

(resuelvo bien)

$$\Rightarrow \pi\frac{x^3}{3}\Big|_0^2$$

$$=\pi\frac{8}{3}-0=\frac{8}{3}\pi\,u^3$$

Propiedades de la integral definida:

1) $\int_a^b \big(f(x)+g(x)\big)\,dx = \int_a^b f(x)\,dx + \int_a^b f(x)\,dx$

2) $\int_a^b k f(x)\,dx = k\int_a^b f(x)\,dx$

* 3) $\int_a^c f(x)\,dx + \int_c^b = \int_a^b f(x)\,dx$

4) $\int_a^a f(x)\,dx = 0$

5) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$